Note on `gpt.py` computational workflow.

• RMS norm: normalization of token's embedding vector $\boldsymbol{x} \in \mathbb{R}^d$. This is different from transformer paper's LayerNorm.

$$\text{RMSNorm}(\boldsymbol{x}) = \frac{\boldsymbol{x}}{\text{RMS}(\boldsymbol{x}) + \varepsilon} \ , \ \text{RMS}(\boldsymbol{x}) = \sqrt{\sum_{i=1}^{d} x_i^2} \ , \ \boldsymbol{x} = (x_1, ..., x_d) \ , \ \varepsilon > 0 \ .$$

• Rotary Positional Embedding (RoPE): positional embedding using sin, cos positional tensors. This is different from transformer paper's Positional Embedding.

Input x with shape (B, T, H, D) where $B =$batch size, $T =$sequence length, $H =$number of heads in multi-head attention, $D =$dimension of each head (must be even). Let $\sin = (\sin^{t,i})_{0\leq t\leq T-1, 0\leq i\leq D/2}$, $\cos = (\cos^{t,i})_{0\leq t\leq T-1, 0\leq i\leq D/2}$ be tensors each with shape $(1, T, 1, D/2)$. Then for each batch b, each index $t \in \{0, ..., T-1\}$, each head h, RoPE does the following:

$$(x^{b,t,h,0}, ..., x^{b,t,h,D-1}) \to \begin{cases} \cos^{t,i} x^{b,t,h,i} + \sin^{t,i} x^{b,t,h,i+D/2} & \text{for } 0 \leq i \leq D/2 - 1 \ , \\ -\sin^{t,i-D/2} x^{b,t,h,i-D/2} + \cos^{t,i-D/2} x^{b,t,h,i} & \text{for } D/2 \leq i \leq D - 1 \ . \end{cases}$$

Position information, originally in t, is encoded in such rotation via $\cos, \sin$. Usually we take $\cos^{t,i} = \cos\left(\frac{t}{10000^{2i/D}}\right)$ and $\sin^{t,i} = \sin\left(\frac{t}{10000^{2i/D}}\right)$.

• Causal Self-Attention. The self-attention module with specific causal mask construction.

– Number of kv heads n_kv_head | number of heads n_head.

– self-attention. Input x has shape $(B, T, \_)$, then Q, K, V are from input x, i.e., $Q = W_Q^T X$ etc. They are re-shaped into shape $(B, T, \text{n\_head}, \text{head\_dim})$ for Q and $(B, T, \text{n\_kv\_head}, \text{head\_dim})$ for K, V, respectively. Then apply RoPE to Q, K, apply RMSNorm to Q, K, finally apply $(1, 2)$ transpose $(B, T, H, D) \to (B, H, T, D)$ to Q, K, V;

– kv-cache and masked causal attention.

– final projection.